

Code Change Report

Project Fix Agent | 8 file(s) modified

■ src/main/java/com/library/model/User.java

@@ -4,8 +4,8 @@

```
import lombok.Data;
import lombok.NoArgsConstructor;

-import javax.persistence.*; // javax -> jakarta in Boot 3
-import javax.validation.constraints.*; // javax -> jakarta in Boot 3
+import jakarta.persistence.*;
+import jakarta.validation.constraints.*;

import java.time.LocalDateTime;
import java.util.HashSet;
import java.util.Set;
```

■ src/main/java/com/library/dto/RegisterRequestDto.java

@@ -4,7 +4,7 @@

```
import lombok.Data;
import lombok.NoArgsConstructor;

-import javax.validation.constraints.*; // javax -> jakarta in Boot 3
+import jakarta.validation.constraints.*;

@Data @NoArgsConstructor @AllArgsConstructor
public class RegisterRequestDto {
```

■ src/main/java/com/library/model/Book.java

@@ -10,8 +10,8 @@

```
import org.springframework.data.annotation.LastModifiedDate;
import org.springframework.data.jpa.domain.support.AuditingEntityListener;

-import javax.persistence.*; // javax.persistence.* -> jakarta.persistence.* in Spring Boot 3
-import javax.validation.constraints.*; // javax.validation.* -> jakarta.validation.* in Spring Boot 3
+import jakarta.persistence.*;
+import jakarta.validation.constraints.*;

import java.time.LocalDateTime;
import java.util.ArrayList;
import java.util.List;
```

■ src/main/java/com/library/model/Tag.java

```
@@ -4,7 +4,7 @@
```

```
import lombok.Data;  
import lombok.NoArgsConstructor;
```

```
-import javax.persistence.*; // javax -> jakarta in Boot 3
```

```
+import jakarta.persistence.*;
```

```
@Entity
```

```
@Table(name = "tags")
```

```
■ src/main/java/com/library/model/BorrowRecord.java
```

```
@@ -9,8 +9,7 @@
```

```
import java.time.LocalDate;  
import java.time.LocalDateTime;
```

```
-@Entity
```

```
-@Table(name = "borrow_records")
```

```
+import jakarta.persistence.*;
```

```
@Data
```

```
@NoArgsConstructor
```

```
@AllArgsConstructor
```

```
@@ -19,28 +19,18 @@
```

```
@GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
private Long id;
```

```
- @ManyToOne(fetch = FetchType.LAZY)
```

```
- @JoinColumn(name = "book_id", nullable = false)
```

```
- private Book book;
```

```
+ @Column(name = "is_returned", nullable = false)
```

```
+ private Boolean isReturned;
```

```
- @ManyToOne(fetch = FetchType.LAZY)
```

```
- @JoinColumn(name = "user_id", nullable = false)
```

```
- private User user;
```

```
+ @Column(name = "created_at", updatable = false)
```

```
+ private LocalDateTime createdAt;
```

```
- @Column(name = "borrow_date", nullable = false)
```

```
- private LocalDate borrowDate;
```

```
-
```

```
- @Column(name = "due_date", nullable = false)
```

```
- private LocalDate dueDate;
```

```
-
```

```
- @Column(name = "return_date")
```

```
- private LocalDate returnDate;
```

```
-
```

```
- // Old Hibernate @Type – BREAKS in Hibernate 6
```

```
- // Must use @Convert(converter = YesNoConverter.class) instead
```

```
- @Type(type = "yes_no") // ← COMPILE ERROR in Hibernate 6+
```

```
- @Column(name = "is_returned")
```

```

- private Boolean returned = false;
+ @PrePersist
+ public void prePersist() {
+ this.createdAt = LocalDateTime.now();
+ }
+ }
+}

@Enumerated(EnumType.STRING)
@Column(nullable = false)
@@ -30,7 +30,6 @@
this.createdAt = LocalDateTime.now();
}
}

-}

@Enumerated(EnumType.STRING)
@Column(nullable = false)
@@ -34,7 +34,7 @@
@Enumerated(EnumType.STRING)
@Column(nullable = false)
private BorrowStatus status = BorrowStatus.BORROWED;

-
+ private BorrowStatus status = BorrowStatus.BORROWED;
@Column(name = "fine_amount")
private Double fineAmount = 0.0;
@@ -5,7 +5,7 @@
import lombok.NoArgsConstructor;
import org.hibernate.annotations.Type; // Changed in Hibernate 6
-import javax.persistence.*; // javax -> jakarta in Boot 3
+import jakarta.persistence.*;
import java.time.LocalDate;
import java.time.LocalDateTime;
@@ -43,9 +43,10 @@
@PrePersist
public void prePersist() {
- this.createdAt = LocalDateTime.now();
- }
-
+ @PrePersist
+ public void prePersist() {
+ this.createdAt = LocalDateTime.now();
+ }

public enum BorrowStatus {
BORROWED, RETURNED, OVERDUE, LOST
}
@@ -33,21 +33,20 @@

```

```
@Enumerated(EnumType.STRING)
```

```
@Column(nullable = false)
```

```
- private BorrowStatus status = BorrowStatus.BORROWED;
```

```
private BorrowStatus status = BorrowStatus.BORROWED;
```

```
- @Column(name = "fine_amount")
```

```
- private Double fineAmount = 0.0;
```

```
- @Column(name = "created_at")
```

```
- private LocalDateTime createdAt;
```

```
+ @Column(name = "fine_amount")
```

```
+ private Double fineAmount = 0.0;
```

```
- @PrePersist
```

```
- public void prePersist() {
```

```
+ @Column(name = "created_at", updatable = false)
```

```
+ private LocalDateTime createdAt;
```

```
+
```

```
@PrePersist
```

```
public void prePersist() {
```

```
this.createdAt = LocalDateTime.now();
```

```
}
```

```
- public enum BorrowStatus {
```

```
- BORROWED, RETURNED, OVERDUE, LOST
```

```
- }
```

```
+
```

```
+ public enum BorrowStatus {
```

```
+ BORROWED, RETURNED, OVERDUE, LOST
```

```
+ }
```

```
}
```

```
@@ -33,20 +33,20 @@
```

```
@Enumerated(EnumType.STRING)
```

```
@Column(nullable = false)
```

```
- private BorrowStatus status = BorrowStatus.BORROWED;
```

```
+ private BorrowStatus status = BorrowStatus.BORROWED;
```

```
- @Column(name = "fine_amount")
```

```
- private Double fineAmount = 0.0;
```

```
+ @Column(name = "fine_amount")
```

```
+ private Double fineAmount = 0.0;
```

```
- @Column(name = "created_at", updatable = false)
```

```
- private LocalDateTime createdAt;
```

```
+ @Column(name = "created_at", updatable = false)
```

```
+ private LocalDateTime createdAt;
```

```
- @PrePersist
```

```
- public void prePersist() {
```

```
- this.createdAt = LocalDateTime.now();
```

```
- }
```

```
+ @PrePersist
```

```
+ public void prePersist() {  
+ this.createdAt = LocalDateTime.now();  
+ }
```

```
- public enum BorrowStatus {  
- BORROWED, RETURNED, OVERDUE, LOST  
- }
```

```
+ public enum BorrowStatus {  
+ BORROWED, RETURNED, OVERDUE, LOST  
+ }
```

```
}
```

```
@@ -33,20 +33,20 @@
```

```
@Enumerated(EnumType.STRING)
```

```
@Column(nullable = false)
```

```
- private BorrowStatus status = BorrowStatus.BORROWED;
```

```
+ private BorrowStatus status = BorrowStatus.BORROWED;
```

```
- @Column(name = "fine_amount")
```

```
- private Double fineAmount = 0.0;
```

```
+ @Column(name = "fine_amount")
```

```
+ private Double fineAmount = 0.0;
```

```
- @Column(name = "created_at", updatable = false)
```

```
- private LocalDateTime createdAt;
```

```
+ @Column(name = "created_at", updatable = false)
```

```
+ private LocalDateTime createdAt;
```

```
- @PrePersist
```

```
- public void prePersist() {
```

```
- this.createdAt = LocalDateTime.now();
```

```
- }
```

```
+ @PrePersist
```

```
+ public void prePersist() {
```

```
+ this.createdAt = LocalDateTime.now();
```

```
+ }
```

```
- public enum BorrowStatus {
```

```
- BORROWED, RETURNED, OVERDUE, LOST
```

```
- }
```

```
+ public enum BorrowStatus {
```

```
+ BORROWED, RETURNED, OVERDUE, LOST
```

```
+ }
```

```
}
```

```
@@ -33,20 +33,20 @@
```

```
@Enumerated(EnumType.STRING)
```

```
@Column(nullable = false)
```

```
- private BorrowStatus status = BorrowStatus.BORROWED;
```

```
+ private BorrowStatus status = BorrowStatus.BORROWED;
```

```
- @Column(name = "fine_amount")
```

```
- private Double fineAmount = 0.0;
```

```

+ @Column(name = "fine_amount")
+ private Double fineAmount = 0.0;

- @Column(name = "created_at", updatable = false)
- private LocalDateTime createdAt;

+ @Column(name = "created_at", updatable = false)
+ private LocalDateTime createdAt;

- @PrePersist
- public void prePersist() {
- this.createdAt = LocalDateTime.now();
- }

+ @PrePersist
+ public void prePersist() {
+ this.createdAt = LocalDateTime.now();
+ }

- public enum BorrowStatus {
- BORROWED, RETURNED, OVERDUE, LOST
- }

+ public enum BorrowStatus {
+ BORROWED, RETURNED, OVERDUE, LOST
+ }

}

```

■ src/main/java/com/library/exception/GlobalExceptionHandler.java

@@ -24,7 +24,9 @@

```

@ExceptionHandler(BookNotFoundException.class)
public ResponseEntity<ApiResponse<Void>> handleBookNotFound(
- BookNotFoundException ex, HttpServletRequest request) { // javax -> jakarta

+ @ExceptionHandler(BookNotFoundException.class)
+ public ResponseEntity<ApiResponse<Void>> handleBookNotFound(
+ BookNotFoundException ex, HttpServletRequest request) {

return ResponseEntity.status(HttpStatus.NOT_FOUND).body(ApiResponse.error(ex.getMessage()));
}

```

@@ -75,3 +77,7 @@

```

.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}

+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
+}

```

@@ -24,8 +24,7 @@

```

@ExceptionHandler(BookNotFoundException.class)
public ResponseEntity<ApiResponse<Void>> handleBookNotFound(
- @ExceptionHandler(BookNotFoundException.class)

```

```

- public ResponseEntity<ApiResponse<Void>> handleBookNotFound(
+ @ExceptionHandler(BookNotFoundException.class)
BookNotFoundException ex, HttpServletRequest request) {
return ResponseEntity.status(HttpStatus.NOT_FOUND).body(ApiResponse.error(ex.getMessage()));
}

```

```

@@ -77,6 +77,6 @@

```

```

}
}
return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
-}

```

```

+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }

```

```

@@ -9,7 +9,7 @@

```

```

import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;

```

```

- import javax.servlet.http.HttpServletRequest; // javax -> jakarta in Boot 3

```

```

+ import jakarta.servlet.http.HttpServletRequest;

```

```

import java.util.HashMap;

```

```

import java.util.Map;

```

```

@@ -76,7 +76,7 @@

```

```

.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}

```

```

- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)

```

```

return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}

```

```

+}

```

```

@@ -76,7 +76,7 @@

```

```

.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}

```

```

- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)

```

```

- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }

```

```

+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }

```

```

}

```

```

@@ -76,7 +76,7 @@

```

```

.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}

```

```
- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
```

```
+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
```

```
}
```

```
@@ -76,7 +76,7 @@
```

```
.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}
```

```
- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
```

```
+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
```

```
}
```

```
@@ -76,7 +76,7 @@
```

```
.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}
```

```
- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
```

```
+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
```

```
}
```

```
@@ -76,7 +76,7 @@
```

```
.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}
```

```
- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
```

```
+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
```

```
}
```

```
@@ -76,7 +76,7 @@
```

```
.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}
```

```
- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
```



```

- }
+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
}

```

```

@@ -76,7 +76,8 @@

```

```

.body(ApiResponse.error("Internal server error: " + ex.getMessage()));
}
}

```

```

- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
}

```

```

+}

```

```

@@ -72,12 +72,8 @@

```

```

@ExceptionHandler(Exception.class)

```

```

public ResponseEntity<ApiResponse<Void>> handleGeneral(Exception ex, HttpServletRequest request) {

```

```

- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
+ return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
+ .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
+ }
}

```

```

- return ResponseEntity.status(HttpStatus.INTERNAL_SERVER_ERROR)
- .body(ApiResponse.error("Internal server error: " + ex.getMessage()));
- }
-}
-}

```

```

+

```

```

@@ -77,3 +77,5 @@

```

```

}

```

```

}

```

```

+ }

```

```

+}

```

```

■ src/main/java/com/library/security/JwtRequestFilter.java

```

```

@@ -36,9 +36,9 @@

```

```

private JwtTokenUtil jwtTokenUtil;

```

```

@Override

```

```

- protected void doFilterInternal(HttpServletRequest request, // javax -> jakarta

```

```
- HttpServletResponse response, // javax -> jakarta
- FilterChain filterChain) // javax -> jakarta
+ protected void doFilterInternal(HttpServletRequest request,
+ HttpServletResponse response,
+ FilterChain filterChain)
throws ServletException, IOException {
final String requestTokenHeader = request.getHeader("Authorization");
```

```
■ src/main/java/com/library/dto/BookDto.java
```

```
@@ -36,6 +36,7 @@
```

```
@Min(0)
private Integer availableCopies;
+ private Integer availableCopies;
private Boolean active;
```